EU AI Act Ontology – Documentation

1. Scope of the Ontology

This ontology provides a **lightweight proof-of-concept semantic model of the EU AI Act**. The aim is to demonstrate a **rapid ontology engineering workflow supported by AI-assisted concept extraction and human-guided modelling**.

The ontology represents core regulatory concepts including:

- AI systems
- actors involved in the AI lifecycle
- risk categories
- regulatory obligations
- prohibited AI practices

The model focuses on the **risk-based governance structure** of the EU AI Act, where obligations depend on the risk classification of an AI system.

2. AI-Assisted Ontology Development Pipeline

The ontology was developed using a lightweight AI-assisted workflow combining automated concept extraction with manual ontology engineering.

Concept extraction

Relevant summaries and sections of the EU AI Act were analysed to identify key regulatory entities. AI assistance was used to extract candidate concepts which were grouped into categories:

Actors - Provider, Deployer, Distributor, Importer, Market Surveillance Authority
Risk categories - High Risk, Limited Risk, Minimal Risk, Unacceptable Risk
Regulatory obligations - Risk Management System, Data Governance, Human Oversight, Transparency Requirements
Prohibited practices - Social Scoring, Real-time Biometric Identification, Subliminal Manipulation
These concepts formed the initial ontology class hierarchy.

Ontology modelling

The ontology was implemented in **OWL 2 using Protégé**.
Key modelling decisions include:

- Representing **risk categories as classes** linked to AI systems using the property `hasRiskCategory`
- Modelling **actors and obligations as independent entities**
- Representing regulatory relationships using object properties such as `developedBy`, `hasObligation`, and `triggersObligation`
- Adding annotations (`hasArticleReference`) to link ontology concepts to EU AI Act articles

AI tools supported concept discovery and idea generation, while modelling and validation were performed manually.

3. Reuse of Existing Ontologies

To improve interoperability, the ontology reuses established Semantic Web vocabularies:

- **PROV-O** – `prov:Agent` for actors and `prov:Entity` for AI systems
- **W3C Organization Ontology** – `org:Organization` for institutional actors
- **Dublin Core** – `dct:BibliographicResource` for documentation artifacts

These alignments situate the ontology within the Linked Data ecosystem.

4. Logical Modelling

OWL restrictions were used to capture regulatory constraints.
Each AI system must have exactly one risk category:

```
AI_System ⊑ (hasRiskCategory exactly 1 Risk_Category)
```

High-risk systems must trigger regulatory obligations:

```
High_Risk ⊑ (triggersObligation some Obligation)
```

This reflects the EU AI Act's core regulatory logic where **risk classification determines compliance requirements**.

5. Competency Questions

The ontology was designed to answer questions such as:

1. Which AI systems are classified as high-risk?
2. What obligations apply to providers of AI systems?
3. Which AI practices are prohibited under the Act?
4. Which actors are responsible for a specific AI system?

6. Example SPARQL Queries

High-risk AI systems:

```
PREFIX : <http://example.org/eu-ai-act#>
SELECT ?system
WHERE {
  ?system a :AI_System ;
          :hasRiskCategory :High_Risk .
}
```

Obligations of providers:

```
PREFIX : <http://example.org/eu-ai-act#>

SELECT ?provider ?obligation
WHERE {
  ?provider a :Provider ;
            :hasObligation ?obligation .
}
```

Prohibited AI practices:

```
PREFIX : <http://example.org/eu-ai-act#>
SELECT ?practice
WHERE {  ?practice rdfs:subClassOf :Prohibited_AI_Practice .}
```